Work Assignment – Senior Java Engineer

Jeff's Car Rental – Vehicle Rental Cost Summary Module

System Context

Jeff's Car Rental operates a fleet of rental vehicles across Switzerland. To improve transparency and simplify daily accounting, the company wants a small module that calculates what each customer must pay for using a rental vehicle and, as a result, what the company earns per vehicle per day. The goal is not fleet logistics or monitoring, but a clear cost breakdown similar to a printed receipt.

Vehicle types and input data:

- Vehicle type: Compact Van, Large Van, E-Van
- Distance driven (km)
- Energy/fuel used (liters for fuel; kWh for E-Vans)
- Additional services (see below)

Pricing

Base Costs per Vehicle Type

- Compact Van: CHF 0.82 per km
- Large Van: CHF 1.05 per km
- E-Van: CHF 0.68 per km

Energy/Fuel Costs

- Fuel (Compact & Large Vans): CHF 1.95 per liter
- Electricity (E-Van): CHF 0.30 per kWh

Additional Services

- Motorway Day Vignette: CHF 9.00 (required if the customer uses the motorway)
- Gubrist Tunnel Fee: CHF 2.50 per passage (after two paid passages per day, additional passages are free)
- City of Zurich Congestion Fee: CHF 1.00 per km driven inside city limits (does not apply to motorway driving)

Reward Program

- E-Van eco-bonus: If an E-Van drives more than 80 km and uses less than 22 kWh per 100 km, apply a CHF 10.00 eco-bonus (subtract from total).

Your Task

Write a simple program whose output is a clean, human-readable cost summary—similar to a receipt. The daily summary must include:

1. One section per vehicle
2. All cost components (base distance, fuel/electricity, additional services)
3. Eco-bonus when applicable
4. Subtotal for each vehicle
5. A grand total across all vehicles

Input Examples

- E-Van, 95 km, 20 kWh, Motorway Vignette, Gubrist Tunnel x3
- Compact Van, 40 km, 5 liters
- Large Van, 180 km, 15 liters, City Congestion Fee (30 km city driving)

Output Requirements

A daily summary including:

- One section per vehicle
- All cost components
- Additional services(if any)
- Ecobonus (when applicable)-bonus (when applicable)
- Subtotal for each vehicle
- A grand total across all vehicles

Important

- Use only Java SE, at least v17, and JUnit, at least v5
- Test Driven Development (TDD) is a must; automated tests are required
- No GUI and no persistence required—this is a simple calculation module
- If you encounter ambiguity, make a reasonable assumption and document it

Delivery

- Provide source code and tests in a shared repository (GitHub or Bitbucket) or send us a zip file.
- Include clear instructions to build and run tests